

Benjamin Roedell

Solution Architect

Tampa, FL

[LinkedIn](#) [GitHub](#) [GitHub summary](#) [StackOverflow](#) [@BenjaminRoedell](#)

SUMMARY

I am a software architect and engineer. I work across application development, cloud infrastructure, and security, specializing in SOLID, clean code, and unit and integration testing.

I raise delivery quality through automation and advocate for simple, maintainable architecture.

TECHNOLOGIES

Java · Gradle · C# · ASP.NET Core · Redis · JavaScript · TypeScript · Angular · React · Playwright · Powershell · JUnit5 · SQL · Docker · Kubernetes · AWS/Azure · CloudFormation · NestJS · RabbitMQ · MongoDB · Snyk · OAuth

WORK EXPERIENCE

Solution Architect

Sep 2023 – Present

Mad Mobile, Tampa, FL · Restaurant and Retail Software

- Standardize build automation, Snyk security testing, unit and integration tests, code review, release versioning, and deployment automation across old and new applications
- Rewrite ShopCPQ in NestJS, replacing MongoDB queues with RabbitMQ and in-memory timers with Kubernetes jobs
- Make JSDoc type safety and unit test coverage standard for JSRunner implementation repositories
- Guide integration choices across n8n, Prismatic, and NestJS while supporting retail customers such as Knitwell Group and West Marine
- Carry OAuth research for Concierge through to an implementation decision, and steer Concierge toward faster client integration, including Concierge AI
- Move Neo toward reproducible builds and deployments
- Partner with the Cloud team on production ownership, with a clear split between prod and non-prod and explicit access controls
- Support Multi-Location Reporting, Enterprise Solutions, and products including CAKE, MenuPad, and Concierge
- Mentor retail engineers and advocate for simple, maintainable architecture in design decisions

Senior Software Developer (Contractor)

Jan 2019 – Present (part-time since Aug 2023)

Baha'i National Center, Wilmette, IL · Not-For-Profit Religious Organization

- Design automatic API documentation from unit test execution
- Create SAML Identify Provider on existing authentication system
- Develop and maintain Azure Functions, ASP.NET Core/MVC and SOAP services (C#)
- Encourage best practices (e.g. SOLID, clean code, etc) and use of DI for IoC in .NET
- Optimize REST services from controller to database via LINQ and EF Core
- Convert UI pages from jQuery to React and JavaScript to TypeScript
- Consolidate source code repositories and CI/CD solutions for automated testing/linting
- Automate builds using Azure DevOps Build and Release Pipelines
- Integrate ARM templates to streamline deployments
- Alert and troubleshoot using Azure App Insights' KQL (Kusto Query Language)
- Update stored procedures (SQL Server) and write SQL unit tests to verify functionality

Senior Software Developer (Contractor)

Jun 2021 – Sep 2023

Flexible, Tampa, FL · Smart Contract Development Services

- Create Smart Contract using Solidity and HardHat including automated unit tests
- Reduce gas by implementing cloning of Smart Contracts
- Troubleshoot Hyperledger FireFly dynamic creation of supernodes

- Investigate Proof of Concept for Binance Smart Chain real time notifications
- Implement Smart Contracts using both Truffle and Hardhat (Solidity)
- Automate CI/CD using GitHub Actions and AWS CloudFormation for NodeJS apps
- Provide sample PHP for submitting signed token transfer requests to Ethereum

Senior Software Developer

Jul 2019 – Jul 2021

PODS, Clearwater, FL · Moving and Storage

- Decouple legacy WCF based applications from Microsoft AX 2009 to enable smooth upgrade from Microsoft AX (C#)
- Troubleshoot issues related to performance and caching (Redis)
- Track down source code for legacy applications for which exact source could not be identified (C#, Visual Basic)
- Refactor common code into Nuget packages published on Sonatype Nexus via Bitbucket pipelines (DevOps, C#)
- Automate integration tests with on-premise TFS 2017 (Powershell)
- Create templates for Azure DevOps pipeline automation
- Participate in PCI DSS gap remediation

Senior Software Developer

Sep 2015 – Jul 2019

Applicant Insight / NoMoreForms, New Port Richey, FL · Background Investigation and Applicant Processing

- Debug and diagnose obscure race conditions in various environments (JavaScript, Java, SQL)
- Maintain and upgrade legacy Java based systems while launching new systems using .NET (C#, XUnit, Angular, TypeScript)
- Model automated containerized deployment strategies (VirtualBox, VMware Workstation, Docker)
- Integrate credit card / ACH payment system to allow applicants to pre-pay client specified fees (Java, C#)
- Create rule based system for automated appointment of insurance agents (C#)
- Implement unit testing for SQL functions
- Script application deployment using TDD (Powershell, Pester)
- Improve Java coding practices by implementing custom logging, unit testing, and static code analysis (slf4j, junit4, junit5, ErrorProne, SonarQube)
- Create Gradle build scripts for legacy Java applications with custom tasks for local deployment, build versions, and packaging (Java, Gradle, Groovy)
- Tweak legacy apps written with PowerBuilder in PowerScript

Software Architect (Part-Time)

Jul 2015 – Jan 2019

Qualityze Inc., Tampa, FL · Qualityze Systems for Regulated Industries

- Implement file conversion using Salesforce Platform Event and Heroku (Java)
- Research managed package automation to scratch orgs (Salesforce DX, Ant)
- Build Quality System as Salesforce managed package (Apex, VisualForce)
- Architect and implement Salesforce managed package secure feature manager
- Implement PDF manipulation using Java running on Heroku uploading to various salesforce orgs (Java, PdfBox, Apex)
- Notify using unobtrusive messages and Salesforce cometD (WebSocket) messaging
- Create method to perform VisualForce injection to increase flexibility of Salesforce managed package (VisualForce)
- Diagnose slow pages in Salesforce and design method to reduce load time
- Design deployment process around github and lack of automation in Salesforce

Software Developer (Contractor)

Dec 2018 – Jan 2019

iReward.io, Trinity, FL · Blockchain Reward System

- Create blockchain reward system on TRON Network (C#, Typescript, Angular)

Software Engineer

Jan 2013 – Sep 2015

CAE USA Inc, Tampa, FL · Military Training Simulation with Active Security Clearance

- Enhance Ant Contrib cpptasks to generate Visual Studio .sln file and desired .vcproj version to enable source code analysis (Java)

- Deploy and maintained central build automation within secured IT environment (Opscode Chef)
- Implement wrapper over single Magnetic Variation model for multiple languages (Ada, C, C++)
- Create Visual Studio solution wizard to streamline adoption of software best practices (JavaScript)
- Support multiple projects simultaneously; not the norm for recent hires (C++, C, Ada with Linux)
- Write Wireshark dissector for network communication with USMC Tactical Environment (Lua)
- Identify and implement a database to combine multiple lists of software assets (SQL, VBA) and convert database access to MS SQL and ASP.NET MVC (C#)
- Participate in numerous Kaizen workshops focusing on subjects ranging from software development best practices to identifying tools for saving money
- Coordinate and present training for software engineers on Automated Testing, Static Code Analysis, and Software Product Lines (Fitsnesse, GoogleTest, Selenium)
- Develop tool for converting Ant XML build instructions to Visual Studio format (C++)
- Team lead for deployment and institutionalizing internally developed Software Production Line tool

Quality Assurance and Regulatory Affairs Manager /

Dec 2003 – Jan 2013

Software Developer

IsoAid, LLC, Port Richey, FL · Medical Device Manufacturer

- Research solutions for validation, auditing, documentation and manufacturing improvements
- Compare and purchase equipment for sterilization, validation, sterile sealing, and automation
- Maintain and upgrade multiple companywide databases from MS Access to MS SQL Server (VBA)
- Optimize Iodine-125 and Palladium-103 decay and tracking systems in MS Access and Excel (VBA)
- Perform both software and equipment validation on critical systems
- Develop and execute validation protocol for electronic signatures
- Manage simultaneous deadlines for regulatory and quality goals
- Upgrade Quality Assurance System to allow paperless availability
- Submit and receive 510(k) submissions to the FDA
- Submit and receive medical license applications to Health Canada and EU notified body
- Create and execute sterilization validation protocols for both Ethylene Oxide and Steam sterilizers

EDUCATION

Bachelor of Science, Computer Engineering

University of South Florida · 2012

TRAINING

[DEV541: SANS Secure Coding in Java/JEE: Developing Defensible Applications](#)

April 2019 (certified August 2019)

[DEV544: SANS Secure Coding in .NET: Developing Defensible Applications](#)

April 2018 (certified August 2018)

[SANS Core NetWars Experience](#)

April 2018

[OWASP AppSec USA: Hands On Security in DevOps and Application Security Automation Workshop](#)

September 2017

[CS169.1x: Software as a Service \(Ruby\)](#)

June 2014

Software Verification & Validation Strategies (by Noblitt and Rueland)

2010

ISO 14971:2007 Medical devices -- Application of risk management to medical devices

2008

ISO 13485:2003 Medical devices -- Quality management systems -- Requirements for regulatory purposes

2005

ADDITIONAL INFORMATION

TDD: I am a proponent of Test Driven Development. I have written tests in Java, C#, C++ Python, Ruby, JavaScript, and even Powershell and SQL. I developed and performed training in automated testing for over 100 developers.

Application Security: In addition to fixing various common sense vulnerabilities (like unauthenticated access and SQL injection) I have begun applying skills learned at various security related training courses.

Spoken Languages: Native English and Spanish speaker